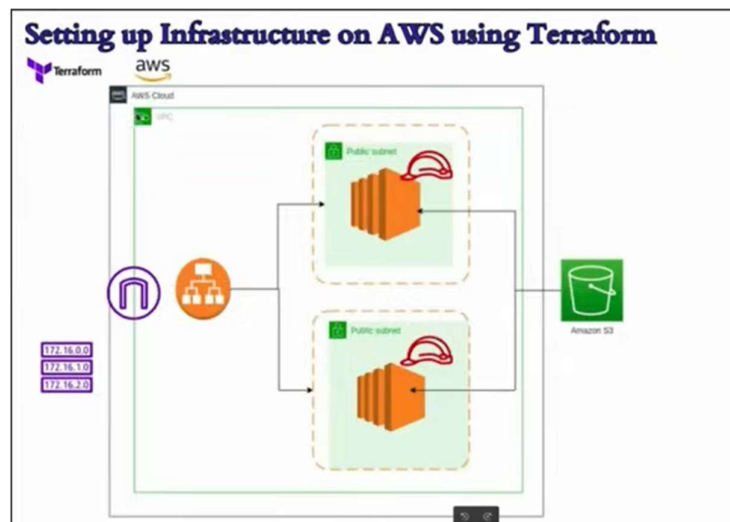


Deploy Portfolio Website on AWS using Terraform with IAM Role-Based S3 Access



The above diagram shows the structure that we are going to create using terraform in AWS console

Step-1: Login to your AWS console with your IAM user and make sure it has all the required access like EC2, S3, Load balancer etc..

Step-2: Verify AWS CLI is installed or not and connect to your AWS console through command prompt using security credentials

Step-3: Install Terraform using below commands

```
sudo apt update
```

```
sudo apt install -y gnupg software-properties-common curl
```

Add HashiCorp GPG key

```
curl -fsSL https://apt.releases.hashicorp.com/gpg | \
```

```
sudo gpg --dearmor -o /usr/share/keyrings/hashicorp-archive-keyring.gpg
```

Add HashiCorp repository

```
echo "deb [signed-by=/usr/share/keyrings/hashicorp-archive-keyring.gpg]  
https://apt.releases.hashicorp.com $(lsb_release -cs) main" | sudo tee  
/etc/apt/sources.list.d/hashicorp.list
```

```
sudo apt update
```

```
sudo apt install terraform
```

The HashiCorp repository provides the Terraform package to APT, while the HashiCorp GPG key allows APT to verify the package's authenticity and integrity. This ensures Terraform is downloaded from a trusted source and hasn't been tampered with.

Verify using `terraform version`

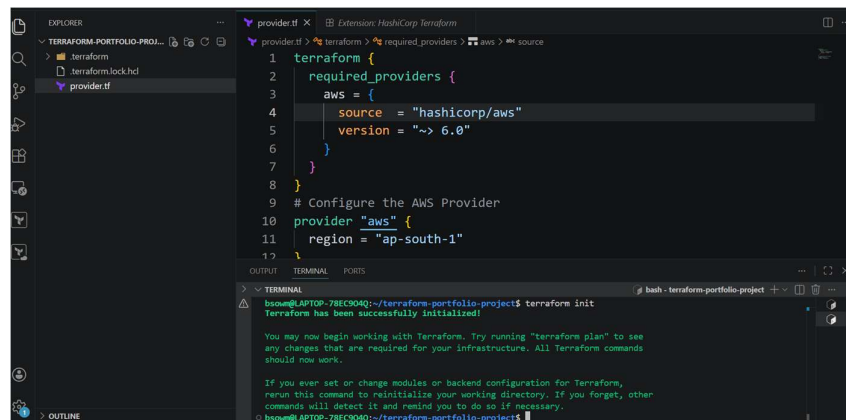
Step-4: Create project directories and open it via VS code

```
mkdir terraform-portfolio-project
```

```
cd terraform-portfolio-project
```

```
code .
```

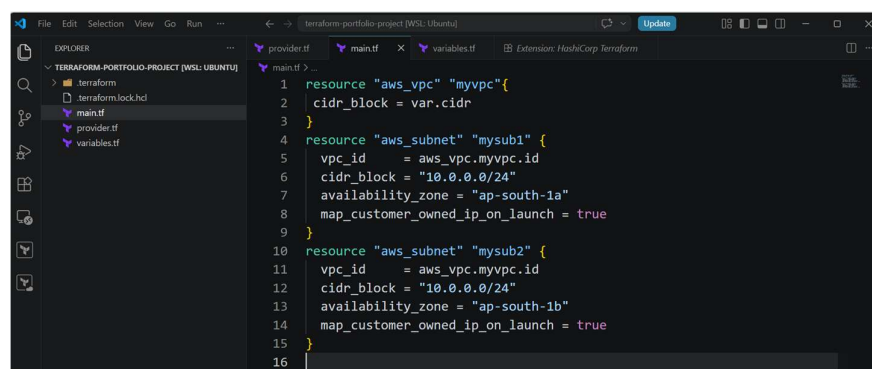
Step-5: After the step-4 it will open VS code and there create provider.tf as we are using aws provider then run `terraform init`(creates some backend files) command to start



Step-6: Use below link to refer documentation

<https://registry.terraform.io/providers/-/aws/6.17.0/docs/resources/vpc>

Step-7: Lets first start with creating VPC and subnets



Step-8: Creating Internet gateway and Route tables and then use below terraform commands to run it.

`terraform validate` – to check if everything is correct

terraform plan – dry run

terraform apply – creates infrastructure in aws console

```
EXPLORER
TERRAFORM-PORTFOLIO-PROJ...
  terraform
    terraform.lock.hcl
  main.tf
  provider.tf
  variables.tf

main.tf
resource "aws_internet_gateway" "myigw" {
  vpc_id = aws_vpc.myvpc.id
}
resource "aws_route_table" "myrt" {
  vpc_id = aws_vpc.myvpc.id
  route = {
    cidr_block = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.myigw.id
  }
}
resource "aws_route_table_association" "myrtba1" {
  subnet_id = aws_subnet.mysub1.id
  route_table_id = aws_route_table.myrt.id
}
resource "aws_route_table_association" "myrtba2" {
  subnet_id = aws_subnet.mysub2.id
  route_table_id = aws_route_table.myrt.id
}
```

Virtual private cloud

Your VPCs

Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv4 CIDR	IPv6 CIDR
Sowmya-VPC	vpc-06a3ae23a40b5095	Available	-	-	Off	10.0.0.0/16	-

Subnets

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR	IPv6 CE
Public-Subnet-1	subnet-0a5923a40b5095a6c	Available	vpc-0940db5ae7130b54c	Off	172.31.0.0/20	-	-
Public-Subnet-2	subnet-0b1653a51a10a98329	Available	vpc-0940db5ae7130b54c	Off	172.31.16.0/20	-	-

Route tables

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
Public-07	rtb-03a23a0273a13139	-	-	Yes	vpc-06a3ae23a40b5095	S99050212593
	rtb-090023a40b5095a6c	-	-	Yes	vpc-0940db5ae7130b54c	S99050212593

Internet gateways

Name	Internet gateway ID	State	VPC ID	Owner
Sowmya-IGW	igw-073023a40b5095a6c	Attached	vpc-06a3ae23a40b5095	S99050212593
	igw-06a3ae23a40b5095	Attached	vpc-0940db5ae7130b54c	S99050212593

Step-9: Creation of security group and s3 bucket

Security Groups

Name	Security group ID	Security group name	VPC ID	Description	Owner
launch-wizard-3	sg-02961185a15865554	launch-wizard-3	vpc-0940db5ae7130b54c	launch-wizard-3 created 2026-06-02T11...	S99050212593
launch-wizard-2	sg-0b4654b7ab39a730e	launch-wizard-2	vpc-0940db5ae7130b54c	launch-wizard-2 created 2026-04-25T11...	S99050212593
myrtba1	sg-0f93a0e473a2a437e	myrtba1	vpc-06a3ae23a40b5095	Managed by Terraform	S99050212593
default	sg-0ba2e29619c019a6d	default	vpc-0940db5ae7130b54c	default VPC security group	S99050212593
launch-wizard-1	sg-0b7a73b73284523a5	launch-wizard-1	vpc-0940db5ae7130b54c	launch-wizard-1 created 2026-04-19T11...	S99050212593
default	sg-0a30a108b0d20a309	default	vpc-0940db5ae7130b54c	default VPC security group	S99050212593

sowmya-terraform-project-bucket-2026

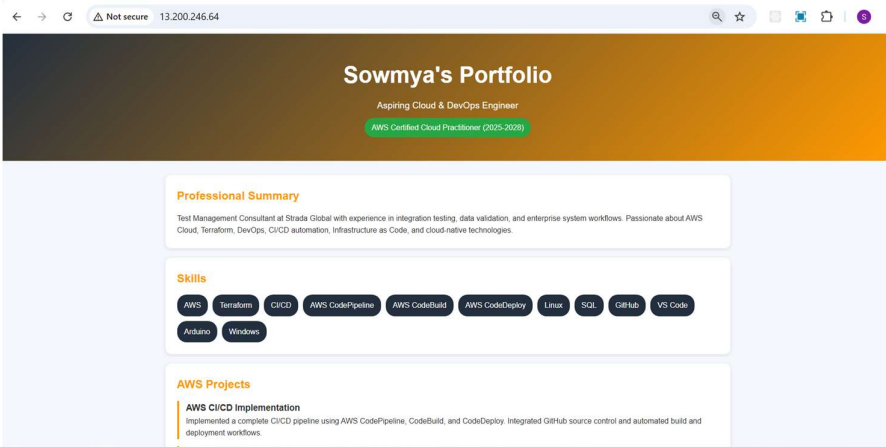
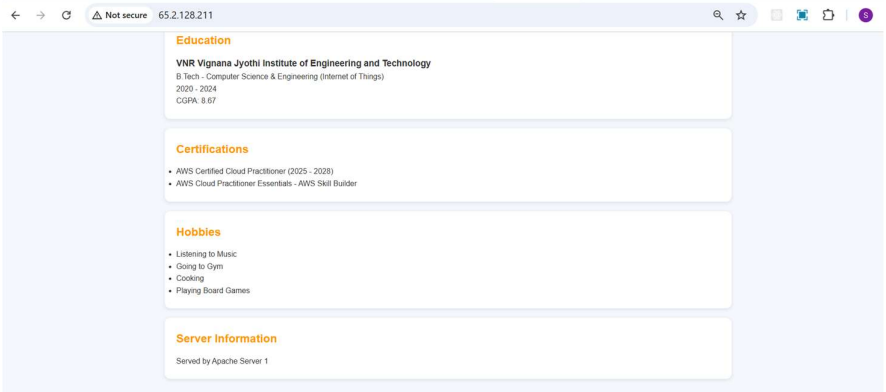
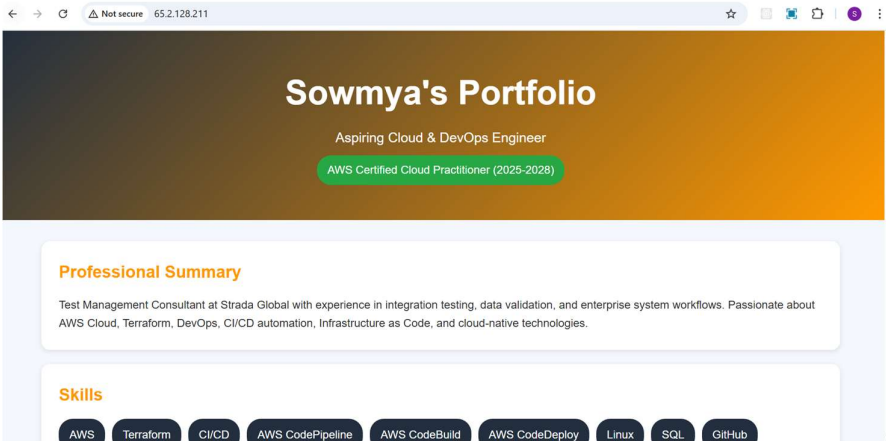
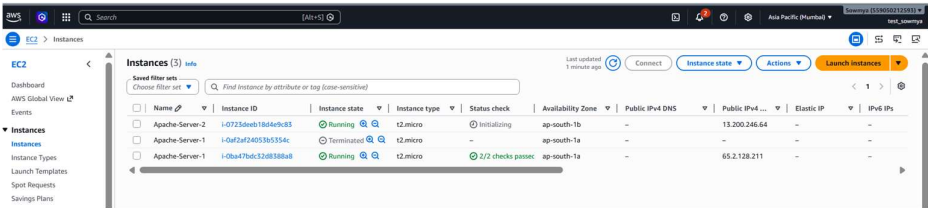
Objects (0)

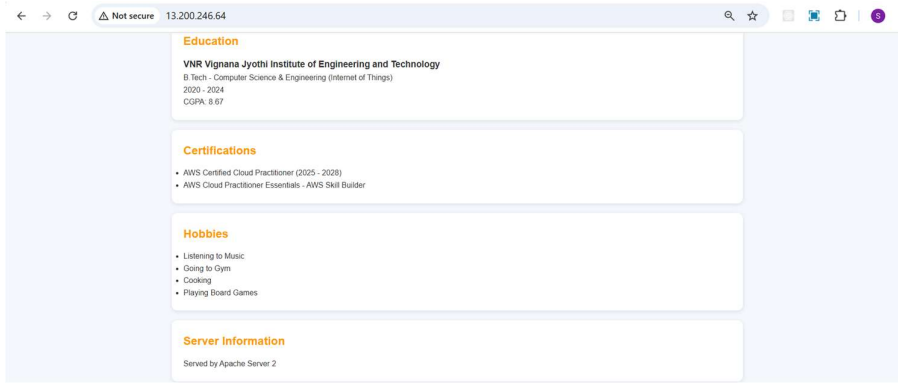
No objects

You don't have any objects in this bucket.

Upload

Step-10: Creation of Instances and attaching IAM role to access S3





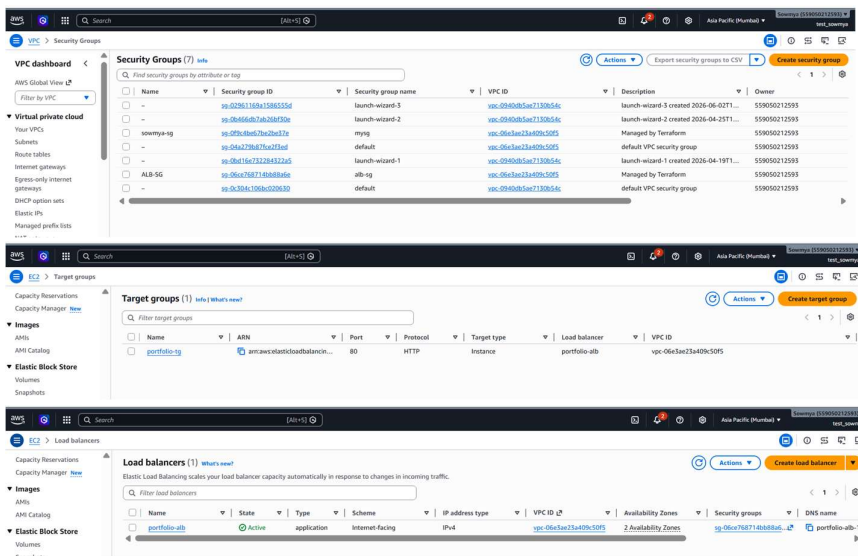
Able to access S3 from EC2 instance

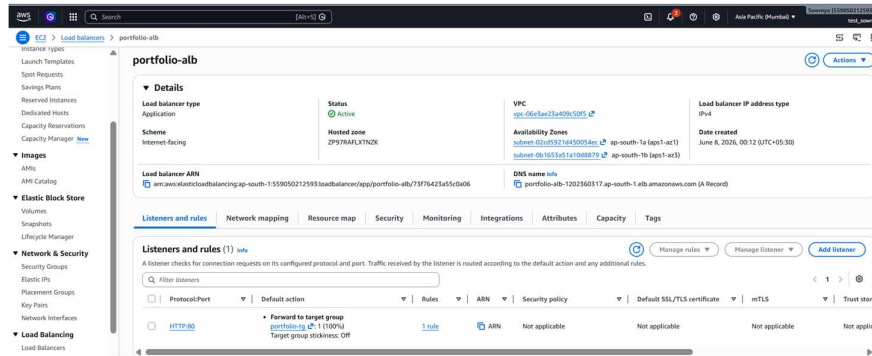
```
[ec2-user@ip-10-0-0-148 ~]$ aws sts get-caller-identity
{
  "UserId": "AROAYEKP5ITY62X4IOJGD:i-0ba47bdc32d8388a8",
  "Account": "559050212593",
  "Arn": "arn:aws:sts::559050212593:assumed-role/EC2-S3-Role/i-0ba47bdc32d8388a8"
}

[ec2-user@ip-10-0-0-148 ~]$ aws s3 ls
2026-04-16 18:51:43 cf-templates-1hdr10lgz8z50-ap-south-1
2026-04-18 20:22:34 codepipeline-ap-south-1-ab35f0035b40-49ca-8d61-78284358f8d2
2026-04-14 12:55:40 sowmya-democli-bucket
2026-06-07 17:20:55 sowmya-terraform-project-bucket-2026
[ec2-user@ip-10-0-0-148 ~]$
```

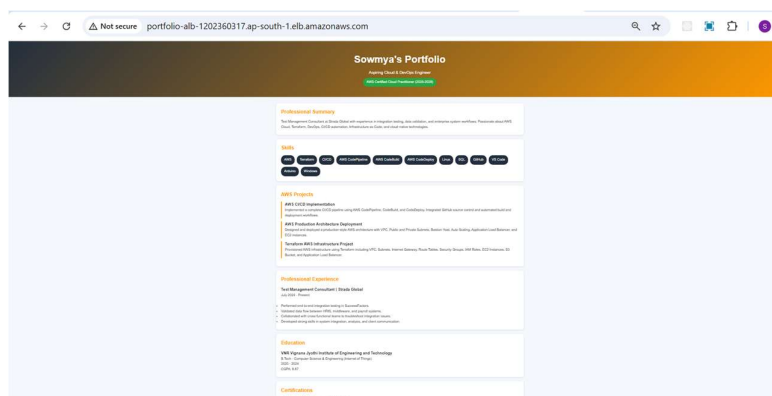
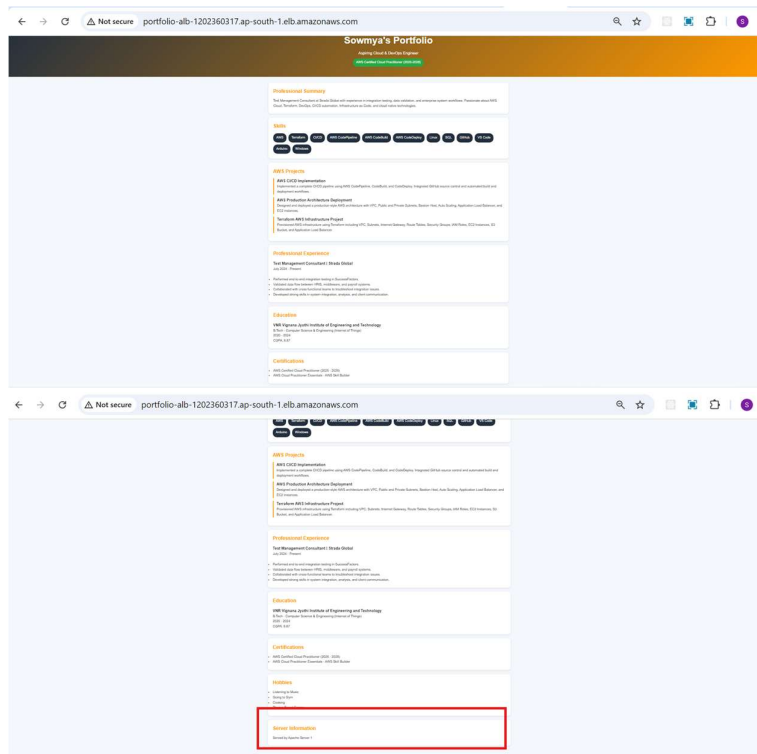
Step-11: Creation ALB security group, Target Group, Load balancer, Registering EC2 instances and listener.

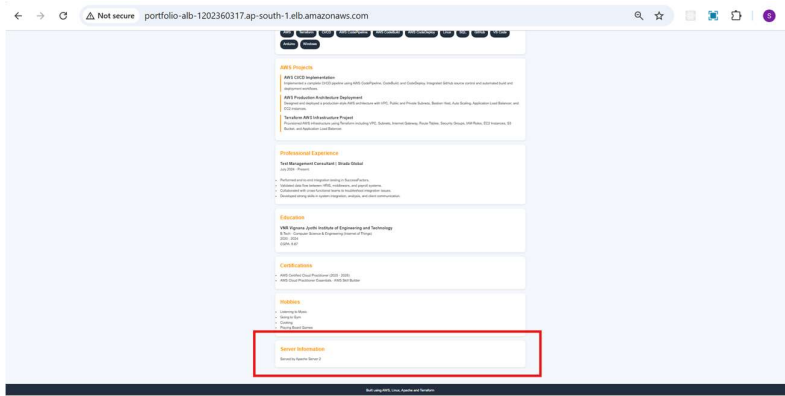
Registering EC2 instances adds them as targets in a Target Group so they can receive traffic from the Load Balancer. A Listener listens on a specific protocol and port (such as HTTP on port 80) and forwards incoming requests to the appropriate Target Group, which then routes traffic to the registered EC2 instances.





Successfully Accessing the Portfolio Webpage





Do not forget to delete, use terraform destroy